Games — problem 13.19

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On any directed graph with one highest and one lowest node, A puts a chip on any node. Than B puts a chip on a an unoccupied node, and so on. If a node is occupied, all lower nodes are forbidden. The player who is forced to place a chip on the highest node loses. Prove that the first player wins if he plays correctly. (*Note* : You are not asked to find the winning strategy. You only need to prove that it exists.)